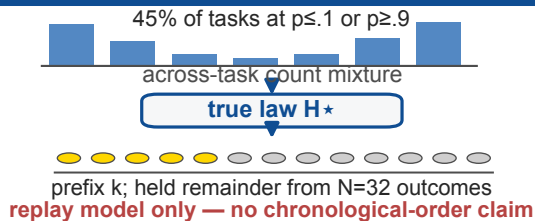


(a) Replay prefix (random; not chronology)

random prefix • sampled without replacement



(b) Oracle H^* versus fitted score

TRUE H^* — oracle theory

$c_{H^*}(x, k) = P(\text{flip} \mid x, k)$
exact conditional • Theorem 1

$\hat{H}_{\text{FIT}}$ — implemented fitted score

$s_{\text{HFIT}}(x, k)$, not c_{H^*}

FIT freeze → CAL EB UCB (Bonferroni)

Theorem 2: marginal replay guarantee

TEST: descriptive; no fitted conditional certificate

(c) Replay quantities + synthetic exploratory drift

OMR $\alpha=.05$ descriptive TEST replay-count reduction

FIXED-HOEF 15.6%

FIXED-EB 46.9%

BAYES-H 80.9% | flip 0.0249

count-replay endpoint; not correctness or operational cost

Synthetic / exploratory / nonconfirmatory drift diagnostic

E3, $\alpha=.05$, $\delta=.15$: BAYES-H 5.2% invalid; FIXED-EB 4.7% valid.
Explored margin $\gamma=.025$: flip 5.0%, reduction 77.2%; exploratory only.

Frozen JSON values; no natural-shift or online-policy claim